

EX. NO: 3**IMPLEMENTATION OF HILL CIPHER****AIM:**

To write a C program to implement the hill cipher substitution techniques.

ALGORITHM:

STEP-1: Read the plain text and key from the user.

STEP-2: Split the plain text into groups of length three.

STEP-3: Arrange the keyword in a 3*3 matrix.

STEP-4: Multiply the two matrices to obtain the cipher text of length three.

STEP-5: Combine all these groups to get the complete cipher text.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {    int a[3][3] =  
{        {6, 24, 1},  
          {13, 16, 10},  
          {20, 17, 15}  
};
```

```
    int b[3][3] = {        {8, 5,  
10},  
          {21, 8, 21},  
          {21, 12, 8}  
};
```

```
    int c[3], d[3];
```

```

char msg[20];

int i, j, t;

printf("Enter Plain Text (3 letters): ");   scanf("%s", msg);

if (strlen(msg) != 3)
{
    printf("Please enter exactly 3 uppercase letters.\n");
return 0;
}

for (i = 0; i < 3; i++)
{
    if (msg[i] >= 'a' && msg[i] <= 'z')
msg[i] = msg[i] - 32;

    c[i] = msg[i] - 'A';
}

for (i = 0; i < 3; i++)
{
    t = 0;      for (j = 0; j < 3;
j++)
    {
        t += a[i][j] * c[j];
    }    d[i] = t % 26;
}

printf("\nEncrypted Cipher Text: ");   for (i = 0; i
< 3; i++)
{
    printf("%c", d[i] + 'A');
}

```

```

for (i = 0; i < 3; i++)
{
    t = 0;
    for (j = 0; j < 3;
j++)
    {
        t += b[i][j] * d[j];
    }
    c[i] = t % 26;
}

printf("\nDecrypted Plain Text: ");
for (i = 0; i
< 3; i++)
{
    printf("%c", c[i] + 'A');
}

return 0;
}

```

OUTPUT:

Output

Clear

```

Enter Plain Text (3 letters): run

Encrypted Cipher Text: XVR
Decrypted Plain Text: RUN

=== Code Execution Successful ===

```

RESULT:

Thus the hill cipher substitution technique had been implemented successfully in C.